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Stability and transition of flow over gapped surfaces in laminar incompressible boundary layers

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The effect of surface gaps on laminar boundary-layer stability and transition is investigated using direct numerical simulations and linear stability analysis of the two- and three-dimensional incompressible Navier–Stokes equations. A systematic range of gap widths, $w/\delta^* \in (0, 130]$, depths, $d/\delta^* \in (0, 4]$, and Reynolds numbers, $Re_{\delta^*} \in [800, 3000]$, is considered, where δ^* is the boundary-layer displacement thickness. In the two-dimensional setting, a comprehensive classification of the long-time dynamics reveals three distinct attracting states: equilibria, limit cycles and chaotic attractors. For equilibrium configurations, the Tollmien–Schlichting waves amplification increment, Δn , is computed using stochastically forced linearized simulations and is found to agree closely with the experimental measurements of Crouch *et al.* (2022). For sufficiently wide gaps, a supercritical Hopf bifurcation gives rise to self-sustained oscillations localized near the downstream cavity edge. Beyond the Hopf bifurcation, a period-doubling cascade provides a route to two-dimensional chaos. The analysis is extended to three dimensions through single-mode spanwise stability analysis. For deep gaps, three-dimensional instabilities arise at smaller gap widths than their two-dimensional counterparts. These results reveal that, for deep-gap configurations, a necessary and sufficient condition for transition is the existence of a two-dimensional periodic orbit that is unstable to three-dimensional perturbations. Such an orbit provides a bypass mechanism to transition, advancing the onset to the downstream edge of the gap. In contrast, purely three-dimensional instabilities are insufficient to induce boundary-layer breakdown. For shallow gaps, transition occurs farther downstream through conventional Tollmien–Schlichting-wave amplification, which is enhanced by the presence of the gap relative to the smooth boundary layer. In the shallow-gap regime, a small number of cases are identified in which a two-dimensional limit cycle remains stable to three-dimensional perturbations.

1. Introduction

2. Numerical setup

The incompressible Navier–Stokes equations are solved with a spectral/*hp* element formulation. In nondimensional form, the governing equations are

$$\begin{aligned}\partial_t \mathbf{u} + \mathbf{u} \cdot \nabla \mathbf{u} &= -\nabla p + \frac{1}{Re} \Delta \mathbf{u}, \\ \nabla \cdot \mathbf{u} &= 0,\end{aligned}\tag{2.1}$$

where $\mathbf{u} = (u, v)$ or $\mathbf{u} = (u, v, w)$ is the velocity field for two- or three-dimensional configurations, respectively, p is the pressure, and Re is the Reynolds number. Throughout the document $\mathbf{x} = (x, y)$ or $\mathbf{x} = (x, y, z)$ denotes the spatial coordinates in two or three dimensions, respectively, where x represents the streamwise coordinate; y , the wall-normal coordinate, and z , the spanwise coordinate. The nondimensionalisation uses the free-stream velocity, u_∞ , as characteristic velocity and the displacement thickness, δ^* , as characteristic lengthscale, which is taken at the upstream gap location of the smooth-wall reference flow. With this choice, $Re \equiv Re_{\delta^*} := u_\infty \delta^* / \nu$, where ν is the kinematic viscosity of the flow. Throughout the study, Re is varied in the range $[800, 3000]$.

Nektar++ solver (Moxey *et al.* 2020) is used to perform all simulations and it is configured with a velocity-correction scheme and a continuous Galerkin projection (Serson *et al.* 2016; Karniadakis *et al.* 1991). Time integration is second-order IMEX, with explicit treatment of nonlinear advection and implicit treatment of viscous terms. Spectral vanishing viscosity is added on top of the equations to stabilize the numerical scheme close to sharp gradients and discontinuities (Kirby & Sherwin 2006).

The computational domain is rectangular, with a cavity of depth d and width w attached to the lower wall to model the surface gap (figure 1). The origin $\mathbf{x} = \mathbf{0}$ is located at the upstream edge of the gap. The dimensions $h = 150\delta^*$, $\ell_i = 100\delta^*$, and $\ell_o = 1000\delta^*$ are selected to ensure a fully developed upstream boundary layer and negligible influence of the downstream and upper boundaries on the region of interest. Boundary conditions are a Blasius profile at the inlet, no-slip on solid walls ($\mathbf{u} = \mathbf{0}$), and far-field conditions at the upper boundary ($\mathbf{u} = \mathbf{u}_\infty(x)$). Dong *et al.* (2014) outflow boundary condition is applied at the outlet to minimize numerical instabilities when energetic vortices pass through the outflow region.

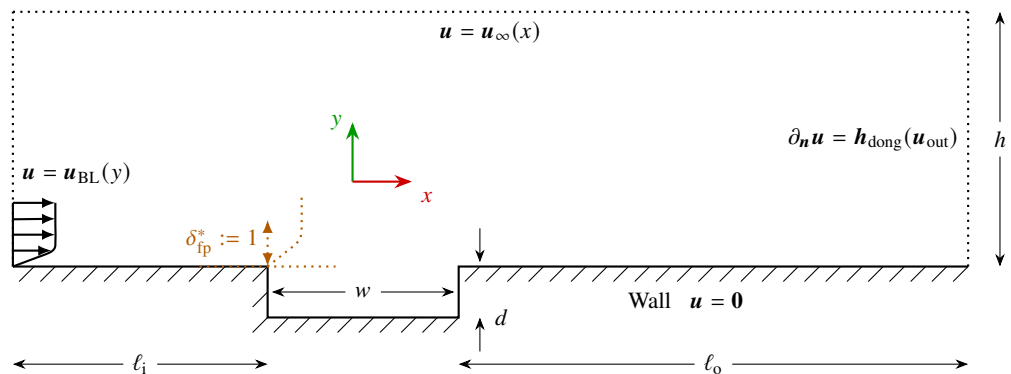


Figure 1. Computational domain and boundary conditions for two-dimensional nonlinear simulations. Dimensions are anisotropically rescaled for clearer visualization of the gap geometry.

For two-dimensional simulations, the domain is meshed with a structured grid of quadrilateral elements in the boundary layer region and unstructured triangular elements in the far-field region. Roughly around 7000 elements are used on each simulation. A polynomial order of $p = 5$ and $p = 4$ is used on every element for the velocity and pressure fields, respectively. Those polynomials lie in the vector space generated by the discrete basis of modified hierarchical Legendre polynomials of the respective dimensions (Karniadakis & Sherwin 2005). This is chosen to ensure that the inf-sup condition is met and thus it prevents numerical instabilities to happen (see Gatica 2014; Olshanskii *et al.* 2021). For three-dimensional simulations, the two-dimensional mesh is extruded in the spanwise direction and periodic boundary conditions are applied. In this dimension, the solution is expanded using 16 complex Fourier modes. The resulting spatial resolution is sufficient to resolve all relevant flow features, as verified by grid-convergence studies.

For global linear stability analysis (LSA), the equations 2.1 are linearized around a base flow $\mathbf{U}$, which is either an equilibrium solution (classical LSA) or a periodic orbit (Floquet analysis). If the base flow is an unstable equilibrium point, the Selective frequency damping method (Jordi *et al.* 2014) is used to damp the unstable eigenmodes and obtain the base flow. Once the base flow is obtained, the linearized equations are solved to obtain the eigenvalues and eigenmodes of the system. These equations read

$$\begin{aligned}\partial_t \mathbf{u}' + \mathbf{U} \cdot \nabla \mathbf{u}' + \mathbf{u}' \cdot \nabla \mathbf{U} &= -\nabla p' + \frac{1}{Re} \Delta \mathbf{u}', \\ \nabla \cdot \mathbf{u}' &= 0,\end{aligned}\tag{2.2}$$

and solutions are seek in the form

$$\mathbf{u}'(x, y, z, t) = \hat{\mathbf{u}}(x, y) \exp(i(\beta z - \omega t)),\tag{2.3}$$

where $\hat{\mathbf{u}}$ is the eigenmode, β is the spanwise wavenumber ($\beta = 0$ for 2D LSA), and ω is the complex eigenvalue. The linearized equations are discretized using the same spectral/ hp element method as for the nonlinear simulations. Since the base flow $\mathbf{U}$ satisfies the non-homogeneous boundary conditions, the linearized equations are solved with homogeneous boundary conditions of the same type. The resulting generalized eigenvalue problem is solved using an modified Arnoldi method (Ohmichi & Yamada 2021).

3. Global stability analysis

3.1. Base flows

For sufficiently small gap depths and widths, the flow asymptotically approaches an equilibrium state, provided that the initial condition lies within basin of attraction of the equilibrium point. The equilibrium flow topology in the boundary layer outside the gap remains close to a Blasius profile, while the flow topology inside the cavity changes strongly with the cavity aspect ratio w/d (figure 2).

For small w/d (figure 2a), a recirculation vortex forms within the cavity beneath the shear layer spanning the gap. This topology is characteristic of an open cavity, for which the separated shear layer remains detached from the cavity floor and the cavity interior is dominated by recirculating flow (Crook *et al.* 2013).

Increasing the aspect ratio to w/d (figure 2b) produces a more elongated, shallow recirculation region. The internal reverse flow is increasingly confined in the wall-normal direction, while the flow still features a recirculation vortex in the vicinity of the downstream edge of the gap. This evolution is consistent with the classical open-to-closed cavity transition, whereby increasing the cavity width-to-depth ratio brings the separated shear layer progressively closer to the cavity floor (Zhang *et al.* 2005; Crook *et al.* 2013).

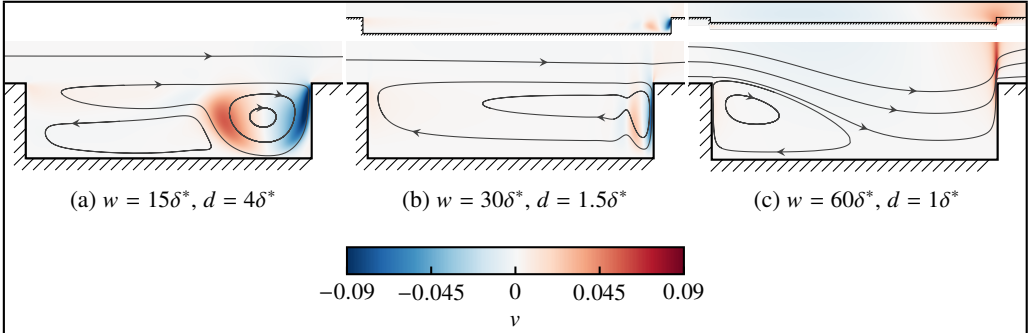


Figure 2. Equilibrium solutions for three representative gap geometries shown using wall-normal-velocity contours and streamlines. All geometries are rescaled to match the aspect ratio of the first case ($w/d = 15/4$) for visual comparison; the original unscaled domain profiles are shown above the last two panels.

Increasing w/d further (figure 2c), the cavity is very shallow and the recirculating motion is confined predominantly to a thin region adjacent to the upstream edge of the gap. In contrast to the deep-cavity case, the outer shear layer exhibits strong streamwise curvature. The resulting structure approaches the closed-cavity topology, for which the separated shear layer reattaches to the cavity floor before leaving the cavity (Zhang *et al.* 2005).

In the limit $w/d \rightarrow \infty$, the flow resembles the superposition of a backward-facing-step (BFS) flow near the upstream edge and a forward-facing-step (FFS) flow near the downstream edge, with recirculation regions confined to the vicinity of each edge and an attached shear layer spanning the gap.

In the limit $w/d \rightarrow \infty$, the two cavity steps become asymptotically separated in the streamwise direction, and the flow may be interpreted locally as a backward-facing-step (BFS) flow in the vicinity of the upstream edge and a forward-facing-step (FFS) flow near the downstream edge. The separated shear layer issued from the upstream step then spans the intervening region and becomes progressively less sensitive to the downstream edge, while the associated recirculation is confined to the vicinity of the upstream step. Near the downstream edge, the shear layer impinges on the wall and is turned back into the external flow, producing a flow topology analogous to that of a forward-facing step.

3.2. Stability of the equilibria

To establish the stability of these equilibrium states, a biglobal stability analysis is performed for both spanwise-uniform ($\beta = 0$) and spanwise-varying ($\beta \neq 0$) perturbations. For the spanwise-varying case (purely three-dimensional), the most unstable eigenvalue is computed for a range of spanwise wavenumbers β and the maximum growth rate is retained. In the simulations a spanwise domain length of $b_1 := 4d$ has been used in all the simulations. According to this choice, and making use of 16 complex Fourier modes, we have a natural set of discrete spanwise wavenumbers $\{\beta_n = 2\pi n/b : n = 0, \dots, 15\}$. Thus stability of spanwise-varying perturbations is assessed by the growth rate

$$\sigma_{3D} = \max_{n=1,\dots,15} \max \{ \Re(\lambda) : \lambda \in \sigma(\mathcal{L}_{w,d,b_n}^{3D}) \}, \quad (3.1)$$

where $\mathcal{L}_{w,d,b_n}$ is the discretized linear operator in equation 2.2 defined in a domain generated with a gap of width w , depth d and spanwise length $b_n := b/n$, and σ denotes the spectrum of the operator. In a similar fashion, the growth rate of spanwise-uniform (two-dimensional) perturbations is defined as

$$\sigma_{2D} = \max \{ \Re(\lambda) : \lambda \in \sigma(\mathcal{L}_{w,d}^{2D}) \}, \quad (3.2)$$

where $\mathcal{L}_{w,d}^{2D}$ is the discretized linear operator in equation 2.2 defined in a two-dimensional domain generated with a gap of width w and depth d . Assessing when the growth rates σ_{2D} and σ_{3D} become positive as a function of w and d allows the identification of the two- and purely-three-dimensional instability boundaries in the $(w/\delta^*, d/\delta^*)$ plane. The frequency of the most unstable mode 2D mode is given by

$$\omega_{2D} = \Im(\arg \max \{ \Re(\lambda) : \lambda \in \sigma(\mathcal{L}_{w,d}^{2D}) \}), \quad (3.3)$$

and equivalently for the three-dimensional case. Figure 3 compares those boundaries together with the experimental by-pass transition boundary observed by Crouch *et al.* (2022) at $Re = 1000$.

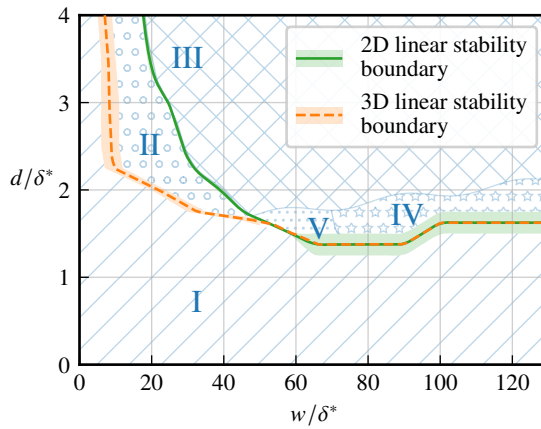


Figure 3. Three-dimensional and two-dimensional linear stability boundaries in the $(w/\delta^*, d/\delta^*)$ plane at $Re = 1000$. Shaded bands indicate interpolation uncertainty obtained from the discrete simulation grid. Dashed brown line represents the experimental boundary beyond which flow transition is observed at the gap location rather than through downstream primary instability mechanisms (Crouch *et al.* 2022). Different regions indicate the corresponding dynamical regimes discussed in the text.

For deep gaps ($d/\delta^* \gtrsim 1.75$), the three-dimensional instability boundary lies at smaller gap widths than the two-dimensional boundary, indicating that spanwise disturbances can destabilize the flow before the two-dimensional equilibrium becomes unstable. For shallow gap ($1.75 \gtrsim d/\delta^* \gtrsim 1.375$), the opposite behaviour is observed, with the two-dimensional instability occurring first. For deep gaps, the two-dimensional boundary closely follows the experimental boundary separating downstream TS-wave transition from transition at the gap location reported by Crouch *et al.* (2022). However, for shallow gaps, it is the three-dimensional stability boundary the one that agrees closely with the experimental boundary. To make this observation more precise, we denote the two linear stability boundaries as $w \mapsto d_{2D}(w)$ and $w \mapsto d_{3D}(w)$, respectively. It is noted that those curves are well-defined in an interval $\mathcal{W}_{2D}, \mathcal{W}_{3D} \subset [0, \infty)$ and that $\mathcal{W}_{2D} \subset \mathcal{W}_{3D}$. In this context, the experimental by-pass transition boundary closely matches the function

$$w \mapsto \begin{cases} d_{3D}(w), & w \in \mathcal{W}_{3D} \setminus \mathcal{W}_{2D}, \\ \max\{d_{2D}(w), d_{3D}(w)\}, & w \in \mathcal{W}_{2D} \cap \mathcal{W}_{3D}. \end{cases} \quad (3.4)$$

define in $\mathcal{W}_{3D}$. It is noted that global flow instability happens after passing through the boundary defined by

$$w \mapsto \begin{cases} d_{3D}(w), & w \in \mathcal{W}_{3D} \setminus \mathcal{W}_{2D}, \\ \min\{d_{2D}(w), d_{3D}(w)\}, & w \in \mathcal{W}_{2D} \cap \mathcal{W}_{3D}, \end{cases} \quad (3.5)$$

This agreement suggests that both two- and three-dimensional instabilities are necessary to by-pass the primary mechanism of transition in the boundary layer (see Section 5).

The boundary curves in figure 3 partition the plane in several regions of interest. Region I corresponds to equilibria that are locally stable to both two- and three-dimensional perturbations. Region II corresponds to equilibria that are locally stable to two-dimensional perturbations but unstable to three-dimensional disturbances. Region III, IV and V correspond to the cases where we already have a two-dimensional instability, with region III and IV being stable to three-dimensional perturbations (their difference will be clarified in Section 4) and region V being unstable to three-dimensional perturbations.

The two-dimensional and purely three-dimensional modes differ in both spanwise structure and temporal character. The two-dimensional mode is spanwise uniform and is associated with the oscillation of the cavity shear layer, whereas the purely three-dimensional mode has finite spanwise wavenumber and produces alternating streamwise structures. The most unstable modes for $(w/\delta^*, d/\delta^*) = (19, 4)$ and $Re = 1000$ are shown in figure 4. The modes are visualized using isosurfaces of the streamwise velocity perturbation u' at a particular time snap.

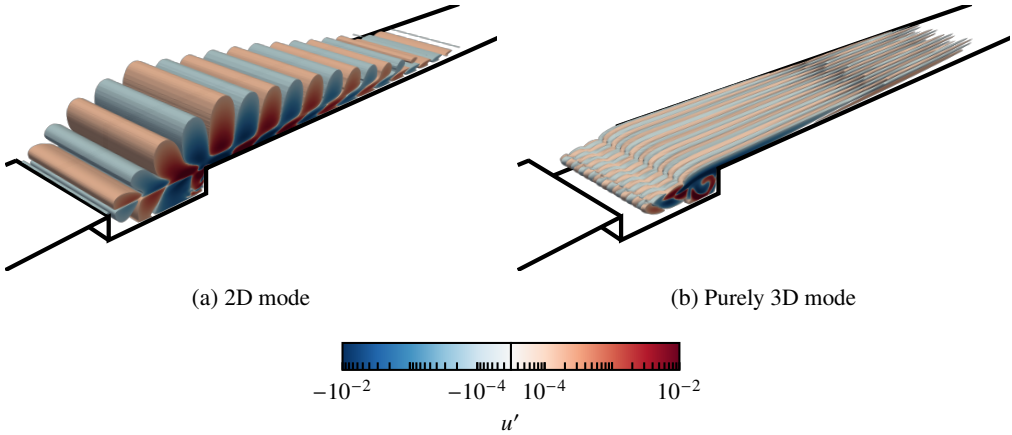


Figure 4. Most unstable two- and purely-three-dimensional mode for the gap configuration $(w/\delta^*, d/\delta^*) = (19, 4)$ at $Re = 1000$. The mode is visualized using isosurfaces of the streamwise velocity perturbation u' in a two-sided logarithmic scale.

The three-dimensional mode develops long streamwise streaks downstream of the gap. An artificial two-sided logarithmic scale is used to reveal the relatively weak magnitude of the streaks compared with the much larger magnitudes of u' inside the cavity. Conversely, the magnitude and isosurfaces of the two-dimensional mode are more intense and their magnitude peaks at the downstream edge of the gap. Regarding the frequency of the modes for the present configuration, $\omega_{3D} \approx 0.0256$, compared with $\omega_{2D} \approx 0.2457$ for the two-dimensional mode.

4. Two-dimensional dynamics

Building on the correspondence between the two-dimensional stability boundary and the experimental transition boundary, we now examine the nonlinear two-dimensional dynamics of the flow over the gapped surface. A systematic DNS parameter sweep is performed over gap depths d/δ^* , gap widths w/δ^* , and Reynolds numbers Re . The two-dimensional dynamical regimes identified at $Re = 1000$ are summarized in figure 5.

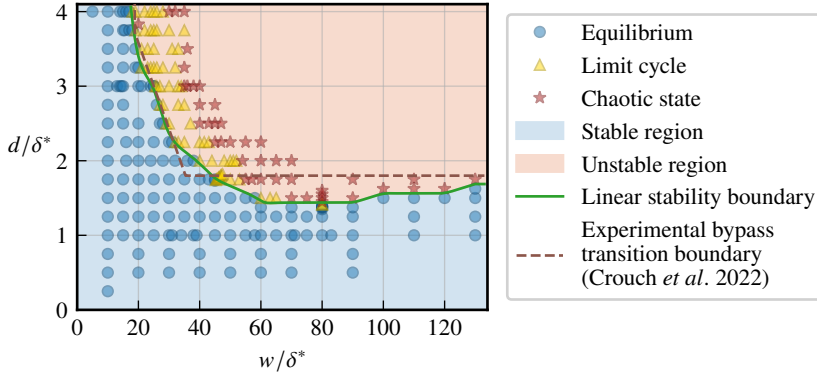


Figure 5. Limit set classification over the parameter space $0 < w/\delta^* \leq 130$ and $0 < d/\delta^* \leq 4$ at $Re = 1000$. Blue circles denote equilibrium states, yellow triangles denote limit cycles, and red stars denote chaotic states. The continuous green line represents the linear stability boundary separating the locally stable and unstable regimes, shaded in blue and red respectively. The dashed brown line represents the experimental boundary beyond which flow transition is observed at the gap location rather than through downstream primary instability mechanisms (Crouch *et al.* 2022).

As said in the previous section, for sufficiently small gap depths and widths, the flow asymptotically approaches an equilibrium state. For sufficiently deep gaps, increasing the width leads to a series of bifurcations that eventually produce chaotic dynamics. The first bifurcation is a Hopf bifurcation, which produces a self-sustained periodic orbit. Further increases in the gap width lead to chaotic dynamics. The 2D dynamics transition to chaos has not been studied in detail in this work, but a few period-doubling bifurcations have been observed in the simulations (see Section 4.2). The shape of the linear stability boundary also deserves attention. As gap depth describes, the critical gap width for the onset of instability increases, until it reaches a minimum in depth, where the influence between both steps is maximum. After that increasing the width stabilizes the system, until eventually reaching the asymptotic limit of two isolated steps.

4.1. TS waves amplification (2D stable regime)

TS waves are the dominant primary convective instability mechanism in two-dimensional laminar boundary layers (Schmid & Henningson 2001; Schlichting & Gersten 2016). However, laminar boundary layers are naturally globally stable to 2D perturbations. Thus, a constant source of noise is required to excite TS waves. To evaluate and corroborate the TS-wave amplification trends reported experimentally by Crouch *et al.* (2022), linearized simulations are performed about the equilibrium base states $\mathbf{U}$ obtained from DNS (blue circles in figure 5 and region I and II in figure 3). A body force $\mathbf{f} = (f_u, f_v)$ is added to the right-hand side of (2.2) to trigger the growth of TS waves. The forcing has a divergence-free Gaussian spatial envelope and is driven in time by zero-mean unit-variance Gaussian

white noise, X_t :

$$f_u(\mathbf{x}, t) = -(y - y_c) \exp(-\alpha|\mathbf{x} - \mathbf{x}_c|^2) X_t, \quad (4.1)$$

$$f_v(\mathbf{x}, t) = (x - x_c) \exp(-\alpha|\mathbf{x} - \mathbf{x}_c|^2) X_t, \quad (4.2)$$

where $\alpha = 100$, $\mathbf{x}_c = (x_c, y_c) = (-95, 0.5)\delta^*$, $|\cdot|$ denotes the Euclidean norm. The spatial envelope therefore confines the forcing to a small circular region of radius approximately $0.5\delta^*$ centred at $\mathbf{x}_c$. Although ambient disturbances in experiments are neither strictly white nor Gaussian, this stochastic model provides a standard and reproducible representation of the external forcing.

The linearized system is integrated forward in time until the initial transient has decayed and a statistically stationary response is reached. Temporal averaging is then used to obtain the stationary root-mean-square (RMS) disturbance amplitude. Following the classical e^n method (Smith & Gamberoni 1956; van Ingen 1956), the amplitude $A(x)$ of a disturbance evolving from a reference position x_0 is written as

$$A(x) = A(x_0) \exp(n_{x_0}(x)), \quad (4.3)$$

where the corresponding local n -factor is

$$n_{x_0}(x) = \log \left(\frac{A(x)}{A(x_0)} \right). \quad (4.4)$$

For brevity, the subscript x_0 is omitted hereafter, such that $n(x) \equiv n_{x_0}(x)$. The disturbance amplitude is defined as the wall-normal L^2 -norm of velocity perturbations:

$$A(x) := \|\mathbf{u}'\|_{L^2(\mathcal{Y}(x))} = \left(\int_{\mathcal{Y}(x)} (u'^2 + v'^2) dy \right)^{1/2}, \quad (4.5)$$

where $\mathcal{Y}(x)$ denotes the wall-normal cross-section at streamwise location x . For the stochastically forced simulations, the corresponding operational amplitude is the RMS value,

$$A(x) := \left(\int_{\mathcal{Y}(x)} \langle u'^2 + v'^2 \rangle dy \right)^{1/2}, \quad (4.6)$$

where $\langle \cdot \rangle$ denotes temporal expectation over the stationary stochastic ensemble. The minimum disturbance amplitude measured upstream of the gap is chosen as the reference $A(x_0)$ for constructing the n curve.

Figure 6 shows the resulting $n(x)$ curves for selected gap configurations. In figure 6a, the gap depth is fixed at $d/\delta^* = 1.5$ and the width is varied in the range $w/\delta^* \in [10, 50]$, while in figure 6b, the gap width is fixed at $w/\delta^* = 40$ and the depth is varied in the range $d/\delta^* \in [0.25, 1.75]$. In all cases the first step of the gap starts at $x = 0$. All the curves exhibit a similar trend: in the gap location, the n -factor increases rapidly, until it peaks at a maximum value just before the downstream edge of the gap. This peak in n is also observed in the experimental results of Crouch *et al.* (2022). Downstream of the gap, the n -factor drops significantly until it recovers a monotonic growth trend similar to the flat-plate reference, with weak variations in the slope. For gaps at constant depth (figure 6a), the BFS has also been included for reference. In this case the BFS produces the maximum amplification, with the gaps approaching this value as the width increases. This aligns with the predictions of Crouch *et al.* (2022). However, it is not clear why after the drop of n just after the second step of the gap, the behaviour of n is slightly different for different gaps. It can be more noticed in figure 6b, where for the deeper gaps ($d/\delta^* > 1.5$), the n -factor struggles to recover as fast as the shallower gaps and, which

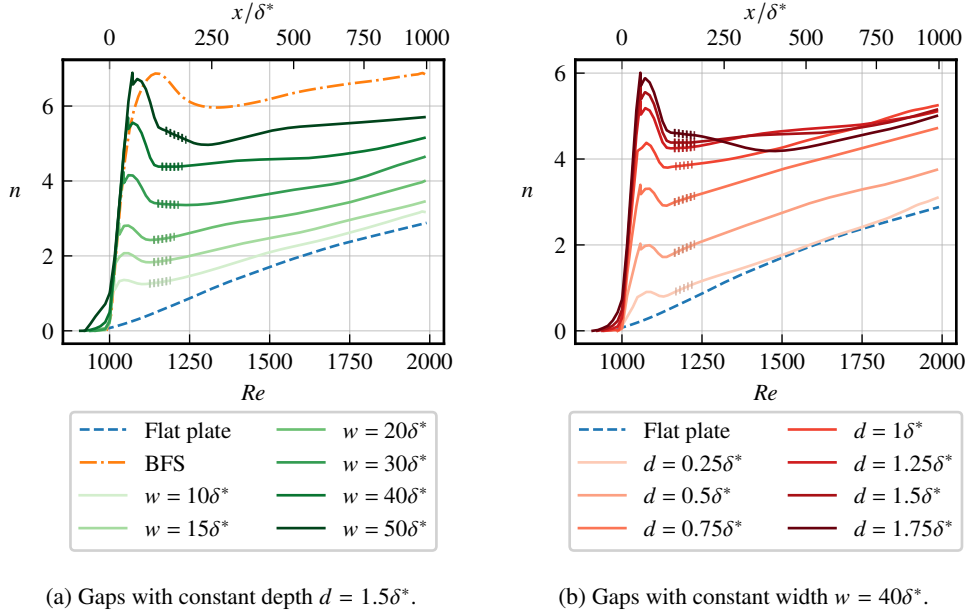


Figure 6. n -factor evolution curves, $n(x)$, along the streamwise direction for different gap configurations at $Re = 1000$. Small markers indicate the streamwise locations used to compute the average Δn values. The flat-plate reference case is shown as a dashed blue line.

in turn produces some intersections between the curves. To examine the origin of these trends, a local spatial stability analysis was performed using the equilibrium base flows and the parallel-flow assumption, $\mathbf{U} = (U(y), 0)$. Under these conditions disturbances are represented by normal modes of the form

$$v'(x, y, t) = \hat{v}(y) \exp(i(\alpha x - \omega t)), \quad (4.7)$$

where $\alpha = \alpha_r + i\alpha_i$ and $\omega = \omega_r + i\omega_i$ are the complex streamwise wavenumber and temporal frequency, respectively, with $\alpha_r, \alpha_i, \omega_r, \omega_i \in \mathbb{R}$. In the temporal formulation, $\alpha = \alpha_r$ is real and ω is the eigenvalue. The resulting Orr–Sommerfeld problem is

$$(D^2 - \alpha_r^2)^2 \hat{v} - iRe [(\alpha_r U - \omega)(D^2 - \alpha_r^2) - \alpha_r U''] \hat{v} = 0, \quad (4.8)$$

where $D \equiv d/dy$. For the cases considered here, the temporal growth rates are converted to spatial growth rates using Gaster's transformation (Gaster 1962; Náráigh & Spelt 2013). This provides the spatial amplification rate $-\alpha_i$ required to construct the spatial local stability boundaries. The base flow profiles are extracted at discrete streamwise locations, excluding the immediate vicinity of the gap corners, where the parallel-flow approximation is not valid. At each x location, the growth rate is evaluated over the relevant frequency range. The frequency is nondimensionalized according to $F = 10^6 \omega_r \nu / u_\infty^2$ (Schmid & Henningson 2001) and the local stability boundary is defined by $-\alpha_i = 0$. Figure 7 shows the resulting neutral stability boundaries in the $(x/\delta^*, F)$ plane for selected gap configurations, together with the flat-plate Blasius reference. The respective Re values based on the displacement thickness are also indicated in another axis for comparison with literature.

Several features of the local stability characteristics explain the observed behaviour of $n(x)$. Within the cavity region, the unstable frequency range broadens and since we are

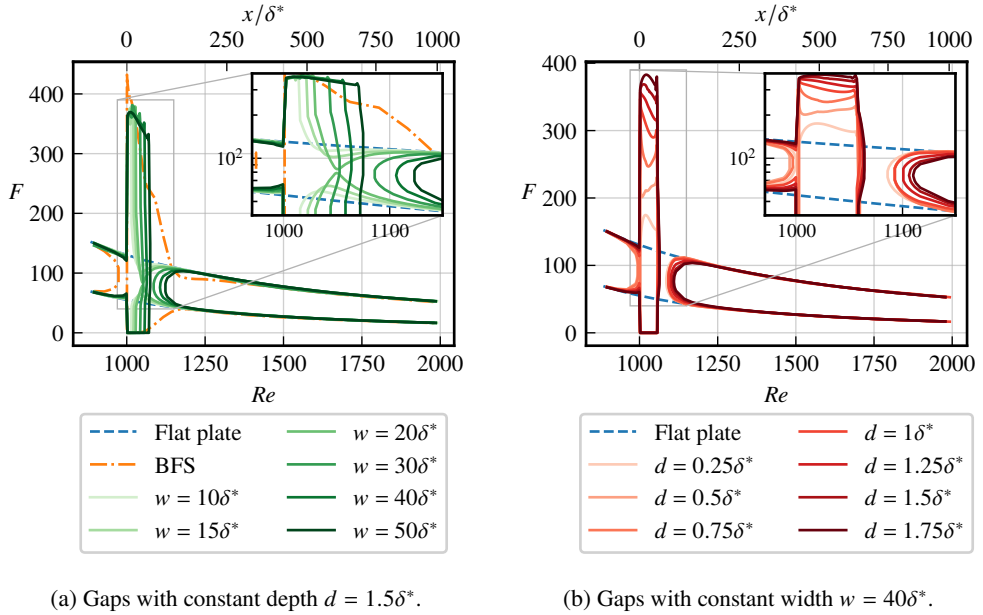


Figure 7. Neutral curves ($-\alpha_i = 0$) of the spatial stability problem for (a) constant gap depth $d = 1.5\delta^*$ with varying width w/δ^* , and (b) constant gap width $w = 40\delta^*$ with varying depth d/δ^* , compared against the flat-plate reference flow.

perturbing the system with white noise, the forcing excites a wider range of frequencies and makes the whole perturbation grow faster. This leads to the enhanced amplification observed in the n curves. In the downstream region, for small gaps such as $(w/\delta^*, d/\delta^*) = (10, 1.5)$ or $(w/\delta^*, d/\delta^*) = (15, 1.5)$, the unstable region gradually returns towards the flat-plate neutral boundary and the n -factor recovers to the flat-plate reference. However, as frequencies have been excited in a very different manner in the cavity region, due to the advective nature of the problem the completely same monotonic growth as in the flat-plate case is not expected even though the stability boundaries match very well at $x/\delta^* \gg w$.

For wider gaps, the unstable region develops two distinct branches separated by a locally stable frequency range. This locally stable region corresponds to the temporary and abrupt reduction in n observed downstream of wide gaps in figure 6 and the difficulty in the recovery of the widest gaps considered in figure 6b.

To compare directly with the experimental results of Crouch *et al.* (2022), the net gap-induced amplification increment of the n -factors, $\Delta n \equiv \Delta n_{\text{gap}} := n_{\text{gap}} - n_{\text{fp}}$, is evaluated relative to the flat-plate reference flow, here denoted as n_{fp} . Since Δn exhibits weak streamwise variation, Δn is computed by averaging over a $50\delta^*$ -long streamwise interval located downstream of the initial amplitude recovery zone (see small dashed markers on top of each amplification curve in figure 6), where the direct effects of the gap and the associated non-parallel flow have decayed. The notation Δn , rather than ΔN , is used throughout to emphasise that specific disturbance-growth histories are compared rather than the global maxima of their respective envelopes.

The metric Δn is evaluated for all parameter configurations that admit a stable steady base flow. To obtain a continuous representation over the stable parameter domain, the computed values are interpolated using Gaussian process regression with a Matérn kernel.

Figure 8 compares the resulting numerical contours (solid blue lines) with the experimental isolines $\Delta n = 1, 2, 3, 4, 5$ reported by Crouch *et al.* (2022) (dashed orange lines).

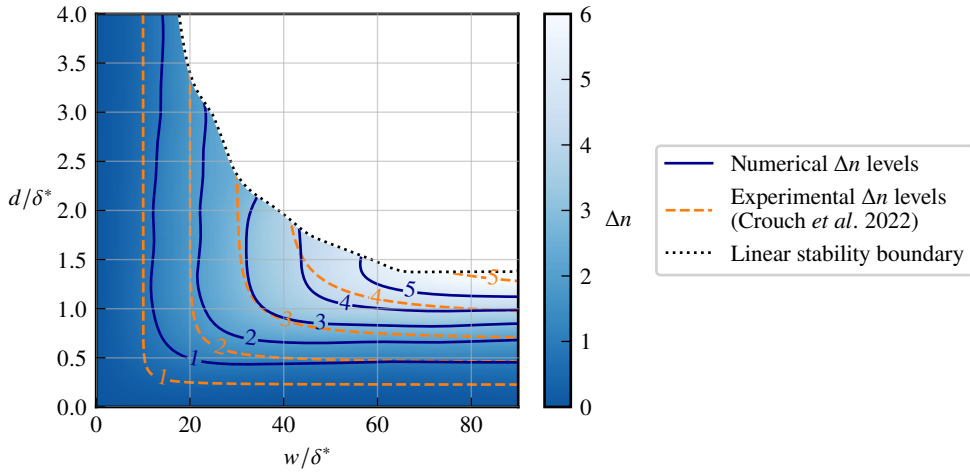


Figure 8. Contour map of Δn on the stable side of the parameter grid $(w/\delta^*, d/\delta^*) \in [0, 90] \times [0, 4]$. Dark blue solid contours indicate the corresponding numerical predictions. Orange dashed contours indicate experimental levels from Crouch *et al.* (2022) for $\Delta n = 1, 2, 3, 4, 5$.

The agreement with experiment is good, with both characteristic trends reproduced. For narrow gaps ($w/\delta^* \lesssim 20$), the isolines are predominantly vertical, whereas for shallow gaps ($d/\delta^* \lesssim 1.5$), they become predominantly horizontal.

The dependence of the primary stability boundary on Reynolds number is shown in figure 9 for $Re \in [800, 3000]$. Increasing Re shifts the Hopf stability boundary towards smaller gap dimensions, indicating that smaller surface imperfections are sufficient to destabilize the flow at higher Reynolds numbers. As w/δ^* increases, the stability boundary approaches the limiting behaviour associated with an isolated FFS, which is inherently more unstable than a backward-facing step (Lanzerstorfer & Kuhlmann 2012*a,b*). At $Re = 1000$, the onset of global instability is found at $d/\delta^* \in (2.75, 3.0)$ for the forward-facing step and at $d/\delta^* \in (3, 3.25)$ for the backward-facing step. A gap width of approximately $w = 500\delta^*$ is sufficient to recover the corresponding asymptotic limit within these bounds, whereas $w = 250\delta^*$ is not.

The Reynolds-number dependence of TS-wave amplification is further illustrated in figure 10, which shows Δn over the stable region at $Re = 800$ and $Re = 3000$. Increasing the Reynolds number substantially increases the amplification associated with a given gap geometry, shifting the Δn contours towards smaller gap dimensions. At $Re = 800$, relatively large gaps are required to produce $\Delta n \approx 3$. At $Re = 3000$, by contrast, even relatively shallow or narrow gaps, such as $w/\delta^* \approx 20$ and $d/\delta^* \approx 1$, can produce $\Delta n > 4$. The stronger TS-wave amplification at higher Reynolds number is consistent with the corresponding shift of the Hopf boundary towards smaller gap dimensions.

4.2. Limit cycles (2D unstable regime)

As the gap width or depth is increased the flow undergoes a first bifurcation (green line in figure 5). This first instability occurs through a supercritical Hopf bifurcation, giving rise to a stable limit cycle localized near the downstream edge of the gap. Further increases in the gap dimensions lead to secondary bifurcations and, eventually, chaotic dynamics. The

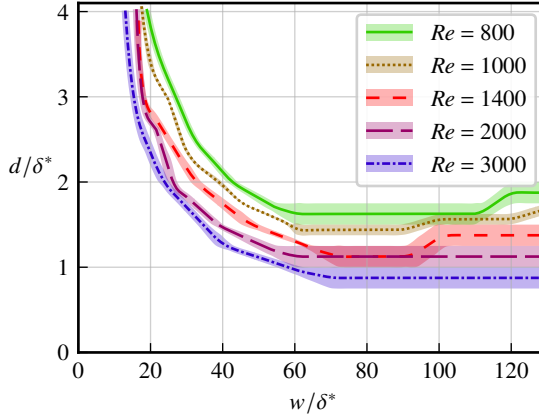
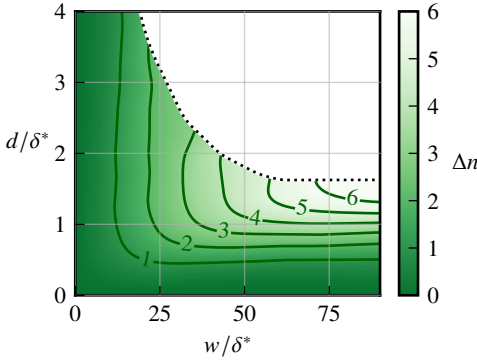
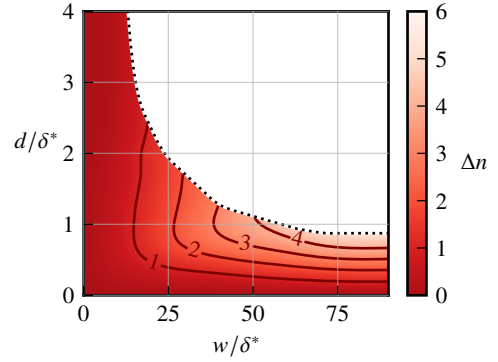


Figure 9. Linear stability boundaries in the $(w/\delta^*, d/\delta^*)$ plane for $Re \in [800, 3000]$. Shaded bands indicate the uncertainty associated with interpolation between the discrete simulation points.



(a) $Re = 800$



(b) $Re = 3000$

Figure 10. Contour maps of numerical predictions of Δn over the stable parameter space $(w/\delta^*, d/\delta^*) \in [0, 90] \times [0, 4]$ for (a) $Re = 800$ and (b) $Re = 3000$.

nature of the secondary bifurcations and transition to two-dimensional chaos is not studied in detail in this work, as it is of no direct relevance because at this stage the flow would already have transition in a three-dimensional setting.

The supercritical nature of the Hopf bifurcation is confirmed by the crossing of a single complex-conjugate pair of eigenvalues through the imaginary axis, while the remaining eigenvalues retain negative real parts. The expected normal-form scaling is further verified by tracking the amplitude of the emerging limit cycle along two continuation paths crossing the Hopf boundary in the $(w/\delta^*, d/\delta^*)$ parameter plane. One path crosses the boundary by varying the gap width at fixed depth, while the other crosses it along an oblique direction in parameter space. Denoting the distance from the bifurcation point along either path by the scalar parameter μ , with $\mu = 0$ at the Hopf boundary and $\mu > 0$ on the unstable side of the equilibrium, the normal form for a supercritical Hopf bifurcation predicts

$$A(\mu) \propto \sqrt{\mu}. \quad (4.9)$$

where A is an appropriate measure of the limit-cycle amplitude. Figure 11 shows the squared L^2 norm of the perturbation wall-normal velocity, $\mathbf{u}' = \mathbf{u} - \mathbf{U}$, as a function of μ along the two paths. The approximately linear dependence of A^2 on μ agrees with the expected square-root scaling of the limit-cycle amplitude.

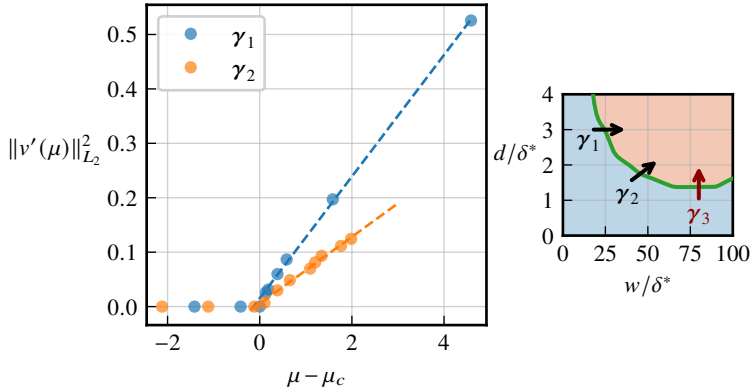


Figure 11. Amplitude evolution of the limit cycle for different crossing locations of the Hopf bifurcation in the $(w/\delta^*, d/\delta^*)$ parameter plane.

Although a single complex-conjugate pair of eigenvalues also crosses the imaginary axis for shallow gaps ($w/\delta^* \gtrsim 75$ and $d/\delta^* \sim 1.5$; γ_3 path in figure 11), the squared perturbation amplitude does not exhibit the expected linear dependence on μ . In this shallow-gap regime, the frequency of the most unstable eigenmode approaches the neutral stability boundary, and the resulting instability becomes increasingly influenced by the spatial growth of the mode. Consequently, its amplitude is not determined solely by the distance from the bifurcation point. Figure 12 shows the evolution of the wall-normal L^2 norm of the perturbation velocity along the streamwise direction for different gap configurations. In the figure, the origin x coordinate refers to the end of all the gaps considered. As mentioned above, the limit cycles are localized near the downstream edge of the gap. It can be seen that for the configurations $w = 60\delta^*$ and $w = 63\delta^*$, the L^2 amplitude considered decays quickly downstream of the gap. As the gap width is increased, the amplitude has a transient growth region downstream of the gap, which indicates that in a particular region of the downstream part of the domain, the frequency of the limit cycle lies just below the neutral stability curve of the domain. Increasing the gap width further, the limit cycle frequency completely lies within the unstable region of the domain and the two mechanisms of instability (global and convective) reinforce each other, leading to a completed destabilized boundary layer.

The fundamental frequency ω of the limit cycles varies with Reynolds number and cavity geometry, as summarized in figure 13.

For narrow gaps ($w/\delta^* \lesssim 60$), the frequencies form three distinct modal branches associated with the cavity geometry and recirculation dynamics. As the gap width increases at fixed depth, the dominant mode of the narrower cavity loses stability and a different periodic state emerges, with a frequency more representative of the larger cavity. The weaker interactions between branches observed across different depths indicate a secondary dependence on the gap geometry. For wide gaps ($w/\delta^* \gtrsim 70$), the limit-cycle frequencies approach the characteristic frequency range of convectively unstable TS waves at the corresponding Reynolds number. The associated modes have longer streamwise scales and lower frequencies.

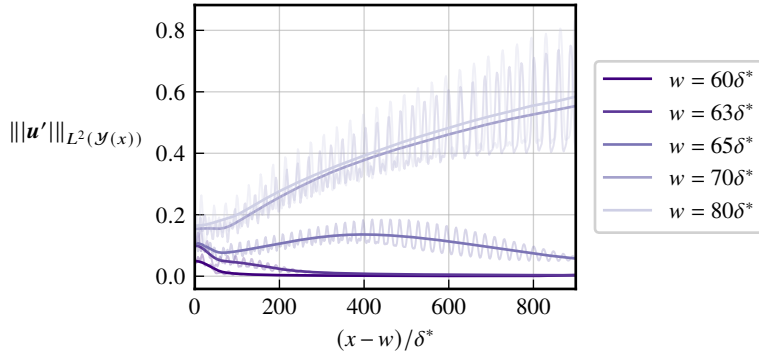


Figure 12. Streamwise distribution of the wall-normal L^2 norm of the magnitude of the perturbation velocity, $|||u'| ||_{L^2(\mathcal{Y}(x))}$, for different gap widths at fixed gap depth $d = 1.5\delta^*$ and $Re = 1000$.

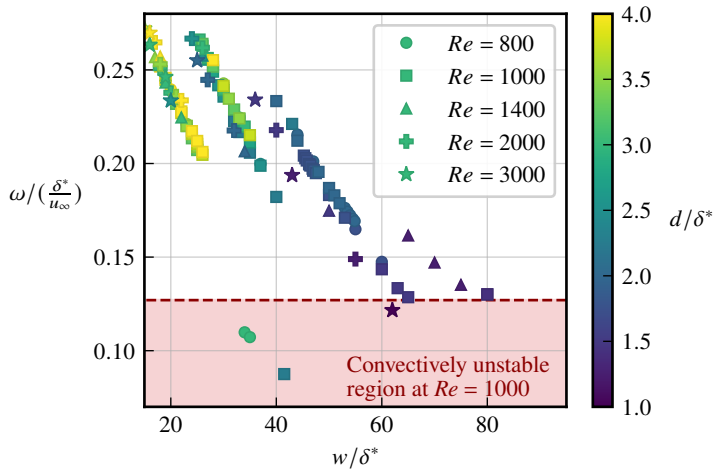


Figure 13. Limit cycle frequencies as a function of gap width w/δ^* for different gap depths and Reynolds numbers. Marker shading indicates gap depth, with darker markers corresponding to shallower gaps and lighter markers corresponding to deeper gaps, and marker shape denotes Reynolds number. The red shaded region indicates the approximately constant frequency range of convectively unstable TS waves at the corresponding Reynolds number in the neighborhood of the gap.

Several isolated points in figure 13 correspond to period-doubling bifurcations, providing a route from periodic to chaotic dynamics. Figure 14 illustrates one such bifurcation for two gap configurations at $Re = 800$. Increasing the gap width from $w/\delta^* = 33$ to 33.5 at $d/\delta^* = 3$ results in a limit cycle with approximately twice the period.

5. Three-dimensional dynamics

Having established the two-dimensional dynamics, we now examine the effect of spanwise perturbations on the stability and transition of the gapped boundary layer. The computational domain is extended in the spanwise direction with periodic boundary conditions. The spanwise extent b is chosen to capture the most unstable spanwise wavelength while avoiding artificial confinement of the three-dimensional structures. The spectral discretization uses 16 complex Fourier modes to resolve the spanwise direction.

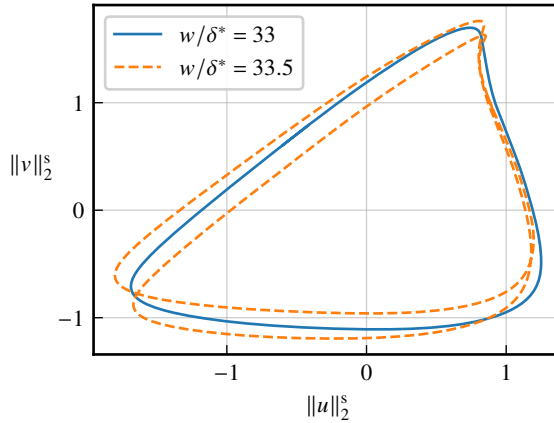


Figure 14. Periodic orbits for the gap configurations $(w/\delta^*, d/\delta^*) = (33, 3)$ and $(w/\delta^*, d/\delta^*) = (33.5, 3)$ at $Re = 800$ projected onto the two dimensional space $(\|u\|_2^s, \|v\|_2^s)$. Here $\|\cdot\|_2^s := (\|\cdot\|_2 - \|\cdot\|_2)/\sigma(\|\cdot\|_2)$ denotes the standardized L^2 norm over the computational domain, with σ denoting the standard deviation of the corresponding time series.

The three-dimensional instability becomes particularly important once the two-dimensional equilibrium has already lost stability. Increasing the gap width beyond the two-dimensional stability boundary leads to a chaotic attractor that develops predominantly near the downstream edge of the gap (figure 15). For deep gaps ($d/\delta^* \gtrsim 2$), this behaviour corresponds to region III in figure 3, whereas for shallow gaps ($d/\delta^* \lesssim 2$) it corresponds to region IV. In both cases, the two-dimensional periodic orbit that emerges after the Hopf bifurcation can become unstable to three-dimensional perturbations. A separate region, denoted region V, contains two-dimensional periodic orbits that remain stable to three-dimensional perturbations; an example is $(w/\delta^*, d/\delta^*) = (60, 1.5)$.

Figure 15 illustrates the transition sequence for the deep-gap configuration $(w/\delta^*, d/\delta^*) = (19, 4)$ at $Re = 1000$. The flow is initialized with a small perturbation in the direction of the most unstable two-dimensional mode (figure 15a). It first evolves towards the corresponding periodic orbit, which is unstable to three-dimensional perturbations. As the instability develops, three-dimensional structures appear both within the cavity and downstream of the gap (figure 15b). The flow, then, subsequently departs from the two-dimensional periodic orbit and develops a chaotic state in the downstream boundary layer (figure 15c). At the later stages, the downstream flow exhibits increasingly complex three-dimensional structures leading to fully developed turbulence (figure 15d). Nevertheless, in the immediate vicinity of the downstream gap edge, the characteristic frequency of the underlying two-dimensional oscillation remains present. This behaviour is consistent with the strong non-TS-wave spectral peak reported experimentally by Crouch *et al.* (2022).

The transition sequence is not restricted to initial perturbations aligned with the most unstable two-dimensional mode. Since the growth rate of the three-dimensional mode is slightly larger than that of the two-dimensional mode, the flow can instead be initialized with a small perturbation in the direction of the most unstable three-dimensional mode. In this case, the three-dimensional disturbance initially follows the linear growth predicted by the unstable eigenmode. As its amplitude increases, nonlinear effects become important and the flow departs from the eigenmode direction, generating finite-amplitude three-dimensional structures that trigger the development and subsequent growth of the two-dimensional mode. The flow then evolves towards a chaotic state similar to that shown in

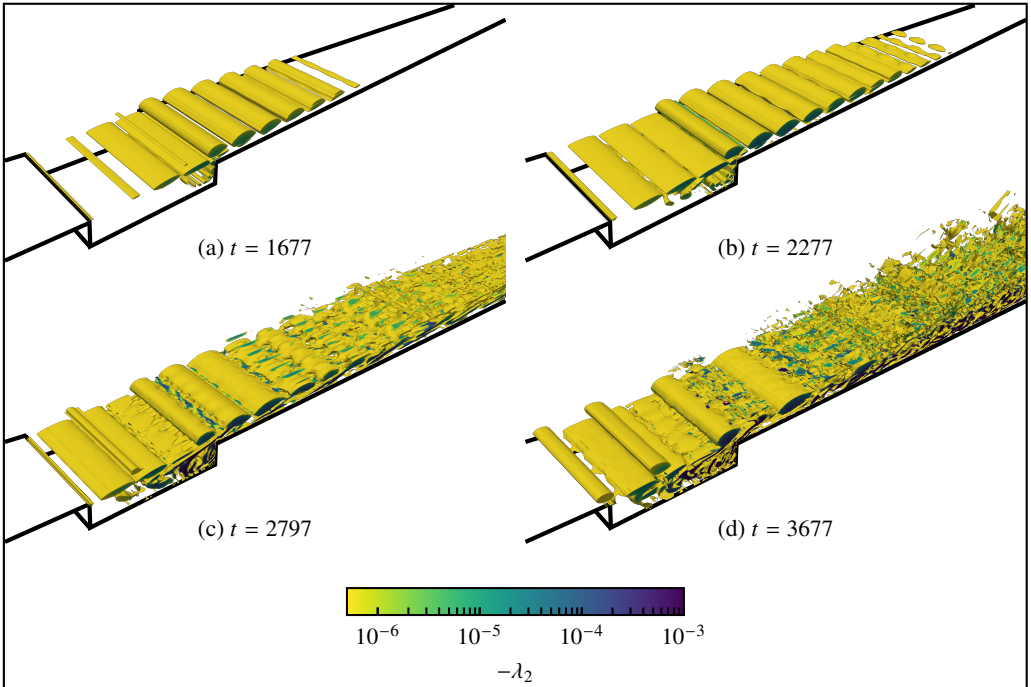


Figure 15. Transition sequence for $w = 19\delta^*$ and $d = 4\delta^*$ at $Re = 1000$. Vortical structures are visualized using λ_2 -criterion isosurfaces. The simulation time corresponding to each state is indicated next to each visualization.

figure 15. These observations show that the three-dimensional instability can provide the perturbation pathway through which transition develops, but that the three-dimensional instability alone is insufficient to produce boundary-layer breakdown. Rather, the bypass mechanism is associated with the interaction between the three-dimensional instability and the two-dimensional oscillatory mode.

To further examine this interaction, a Floquet analysis is performed about the three-dimensionally unstable two-dimensional limit cycle. In contrast to the previous linear stability analysis, in which the base state is a steady equilibrium, the base state for the Floquet analysis is the periodic orbit, $\mathbf{U}_{\text{floq}}(t) = \mathbf{u}_{\text{p.o.}}(t)$, where $\mathbf{u}_{\text{p.o.}}$ denotes the two-dimensional periodic solution.

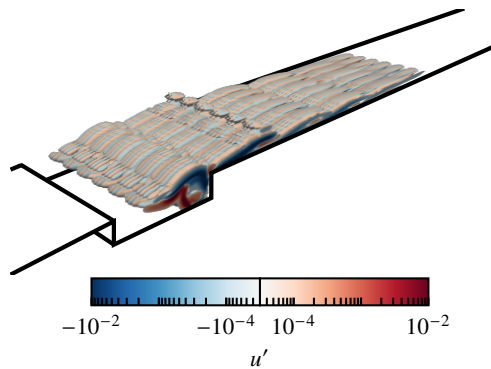


Figure 16. Most unstable floquet mode for the gap configuration with $w = 19\delta^*$ and $d = 4\delta^*$ at $Re = 1000$. Isosurfaces of the streamwise velocity perturbation u' are shown.

Finally, region IV and V in figure 3 corresponds to configurations for which the two-dimensional instability occurs before the three-dimensional instability. The boundary separating regions III and IV, III and V and IV and V has not been resolved in detail. A representative shallow-gap transition in region IV is shown in figure 17. Unlike the deep-gap case, the transition is initiated by the two-dimensional instability and subsequently develops three-dimensional structure farther downstream.

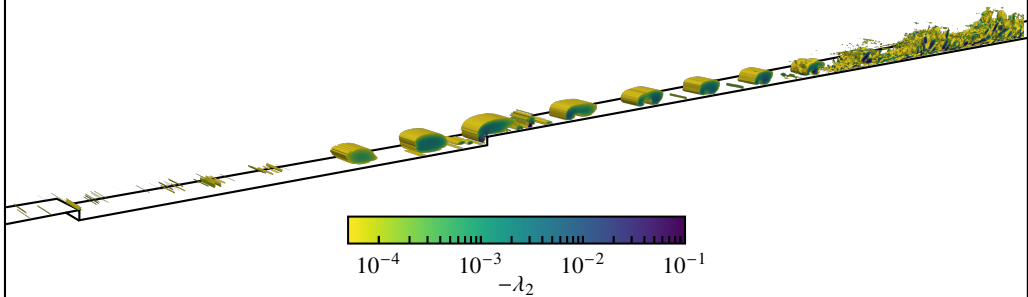


Figure 17. Transition scenario for a gap configuration with $w = 80\delta^*$ and $d = 1.5\delta^*$ at $Re = 1000$. Vortical structures are visualized using λ_2 -criterion isosurfaces.

6. Conclusions

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